Hybrid Bert-Based Fake News Detection Using Linguistic Features And Explainable AI

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Abstract — The rapid circulation of misleading information on digital platforms motivates automated methods for identifying news articles that exhibit patterns associated with fake or real news. This paper presents a hybrid text-classification pipeline that combines contextual representations from BERT with nineteen handcrafted linguistic features. The 768-dimensional BERT [CLS] embedding is concatenated with the linguistic feature vector to form a 787-dimensional representation, which is classified using Logistic Regression. Explainable Artificial Intelligence (XAI) is incorporated using SHapley Additive exPlanations (SHAP) to improve model transparency by identifying the contribution of individual features to predictions and helping users understand the factors influencing classification decisions. A Streamlit application presents predictions and feature-attribution explanations to users. Evaluation on a balanced held-out test set of 1,500 samples achieved 99.2% accuracy and an ROC-AUC of approximately 0.9981. The confusion matrix records 1,488 correct and 12 incorrect predictions. The proposed system combines contextual language representations, linguistic analysis, and explainable predictions to support transparent fake news detection. However, the model identifies learned text patterns and does not independently verify an article's factual truth.

Keywords— Fake news detection, BERT, linguistic features, feature fusion, Logistic Regression, Explainable AI, SHAP

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